



Insight
Analytics

Customer Behaviour Analysis

An interactive Power BI dashboard designed to analyze customer purchasing patterns, revenue trends, and business insights through data visualization and KPI analysis.

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Data Analytics



Project Overview

Key topics and insights from the customer behavior dataset

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About the Dataset

OVERVIEW OF PROJECT SCOPE AND BUSINESS PURPOSE ↗

- **Type:** Retail / E-commerce Dataset
- **Nature:** Structured (Tabular)
- **Level:** Customer Transaction Level
- **Data Mix:**
 - Numerical → Age, Purchase Amount, Previous Purchases
 - Categorical → Category, Payment Method, Season

This dataset is used for **customer behavior analysis + business decision-making.**



Domain Background

In the e-commerce industry, companies track every customer interaction:

- What customers buy
- How often do they buy
- How much do they spend
- Whether they respond to discounts



Why This Data Is Important

This data helps businesses:

- Identify **high-value customers**
- Understand **purchase patterns**
- Optimize **discount strategies**
- Improve **marketing campaigns**
- Increase **customer lifetime value (CLV)**



Business Problem

How can a retail company understand customer purchasing behavior to increase revenue, improve customer retention, and optimize marketing strategies?

Breaking Down the Problem

Low Customer Retention

- Customers are not purchasing frequently

Need to understand:

- Who are frequent buyers?
- Who are one-time buyers?

Ineffective Marketing Campaigns

- Discounts and promo codes may not be effective

Questions:

- Do discounts increase purchases?
- Which customers respond to promo codes

Product Demand Understanding

- Which category performs best?
- Which products generate the highest revenue?

Payment Behavior Insights

- Which payment methods are preferred?
- Your paragraph text
- Are digital payments increasing?

Seasonal Demand Variation

- Does season impact sales?
- Which season drives maximum purchases?

Customer Segmentation Problem

Need to identify:

- High-value customers
- Customers with high spending behavior
- Frequent buyers
- Preferred customer categories and shopping trends

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male/Female/Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Size	Product size (S, M, L, etc.)
Color	Product color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No), Subscription Program
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA – (Exploratory Data Analysis using Python)

05

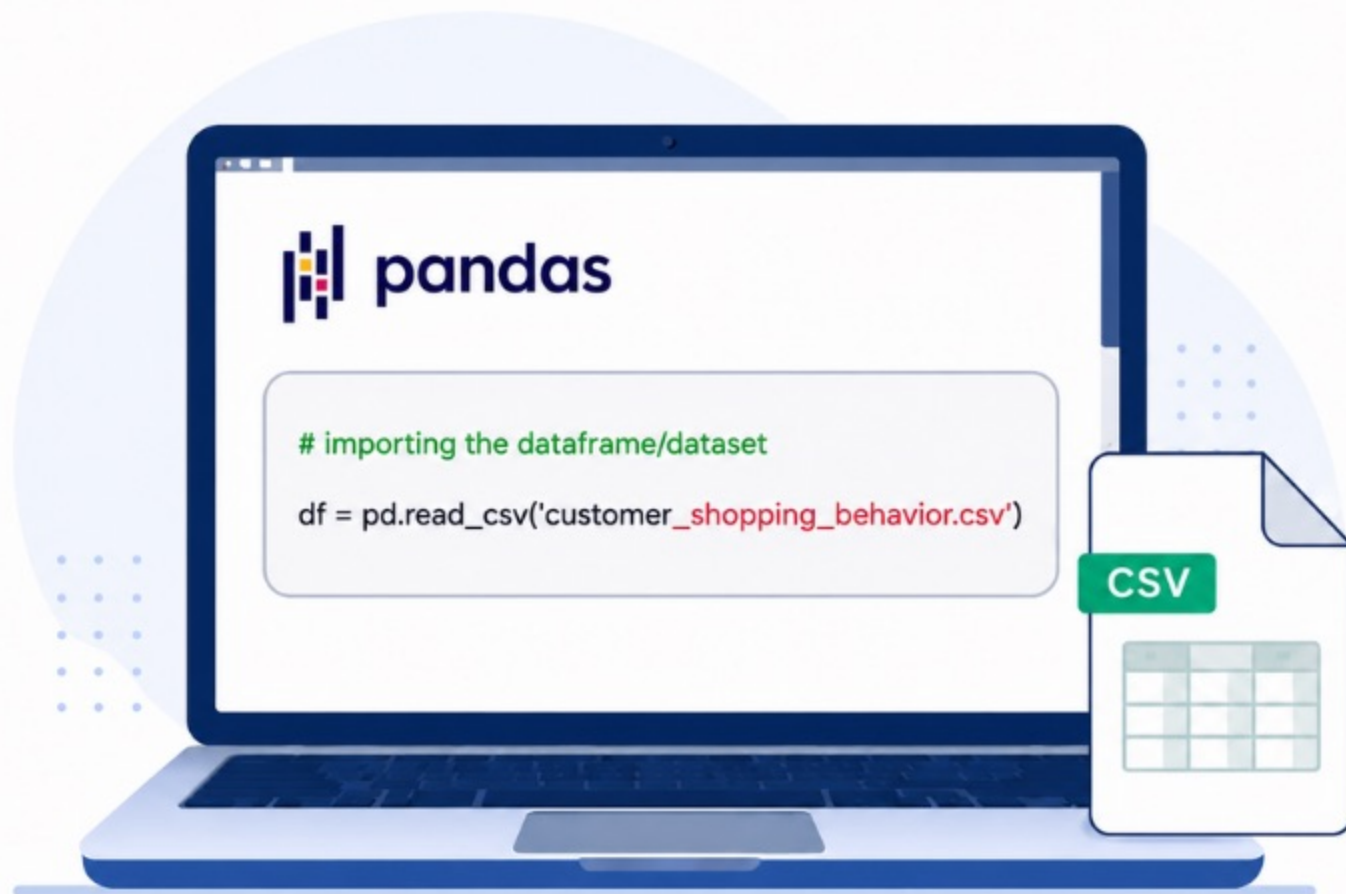
Data Loading

What you did:

- Imported pandas

Why did you do it?

- To bring raw data into a structured DataFrame
- Enables further analysis and transformations



Initial Data Exploration

What you did:

- Viewed data using:
 - `df.head()`
 - `df.tail()`
- Checked dataset summary:
 - `df.describe(include='all')`
 - `df.info()`
- Checked missing values: `df.isnull().sum()`

Why did you do it?

- To understand:
 - Data structure
 - Data types
 - Missing values
- This step helps identify **data quality issues early**

Data Consistency Check (Category Correction)

What you did:

- Checked unique values in:
 - Item Purchased
 - Category
- Found inconsistencies between Item Purchased and Category
- Created a mapping dictionary
- Updated category using:

Why did you do it?

- Categories were inconsistent or incorrect
- Ensures:
 - Accurate grouping
 - Correct business insights

This is a critical **real-world data cleaning step**

Handling Missing Values (Size Column)

What you did:

Handled missing values based on category:

- Electronics → “Not Applicable.”
- Accessories → “Not Applicable.”
- Footwear → “Free Size.”
- Clothing → Filled with mode

Why did you do it?

- Different categories require different business logic
- Example:
 - Electronics don't have size → Not Applicable
 - Clothing → size is important → use mode

This shows **domain understanding**

CUSTOMER BEHAVIOUR ANALYSIS

Category

All

Gender

Female

Male

Other

Discount Applied

No

Yes

Subscription Status

No

Yes



07

5K

Total_customer



30

Unique Items



4

Average Review



994.18K

Total Spend

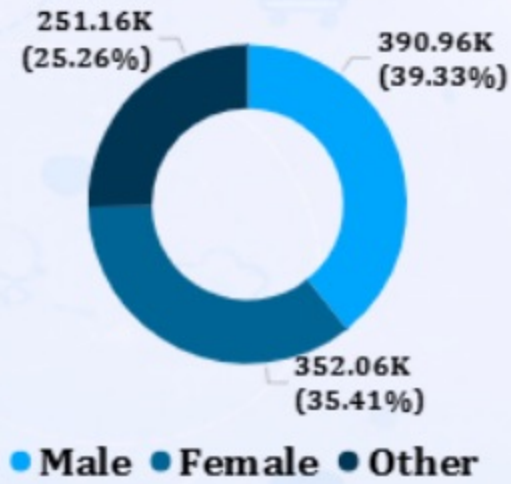


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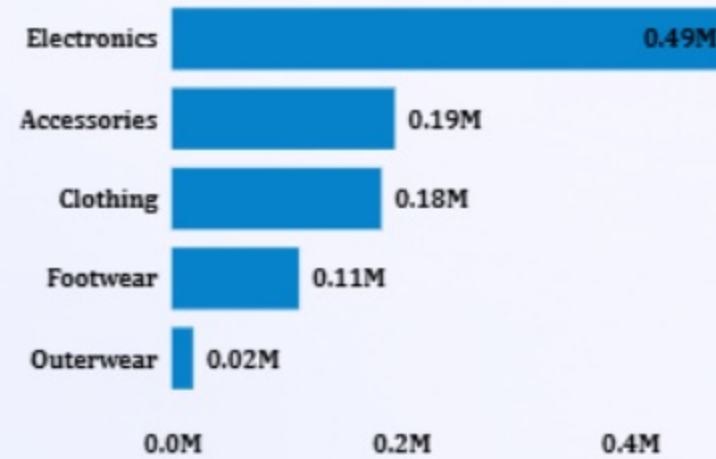
Average Spend



Revenue by Gender



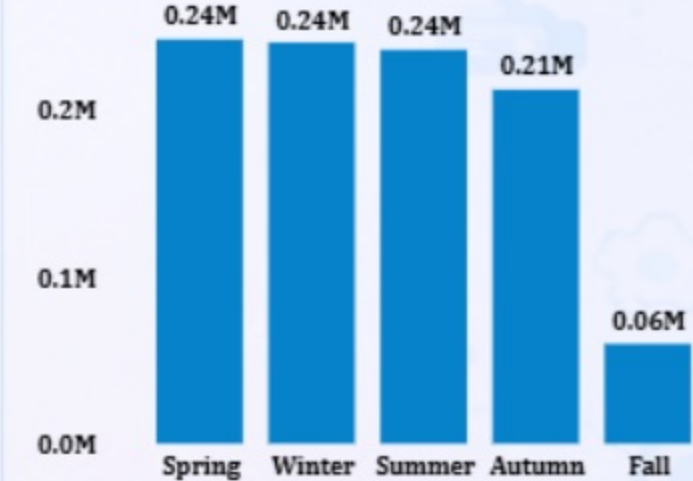
Revenue by Category



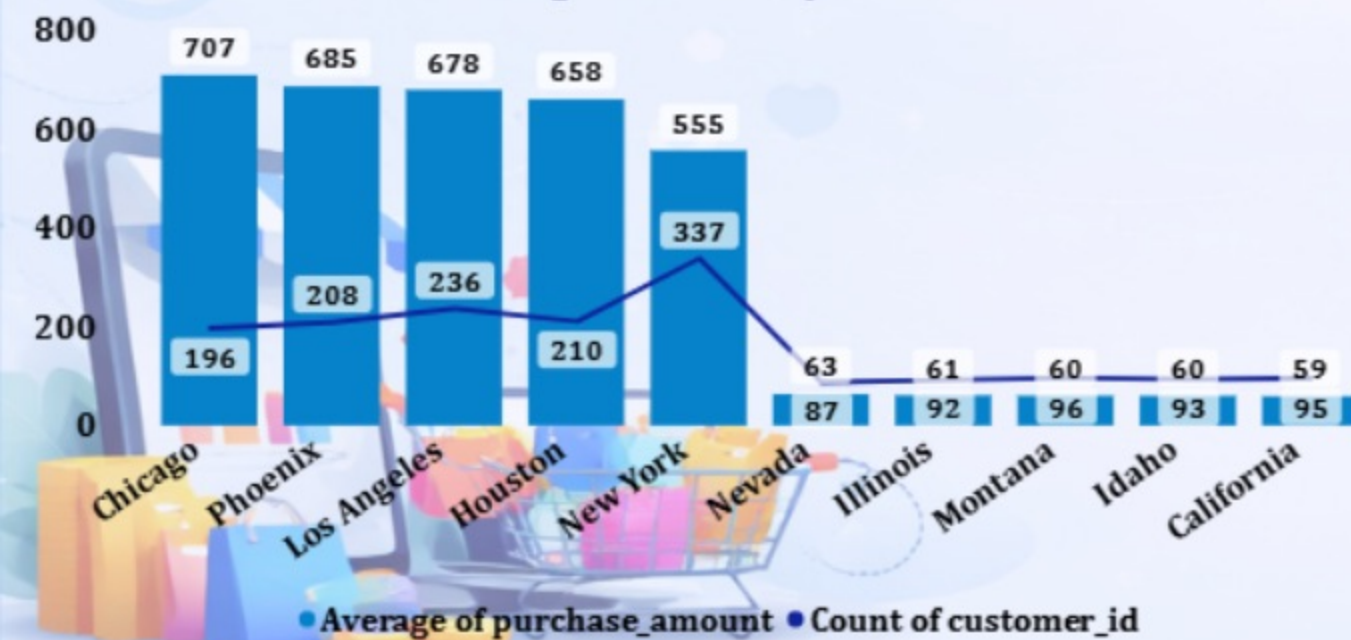
Shipping Type



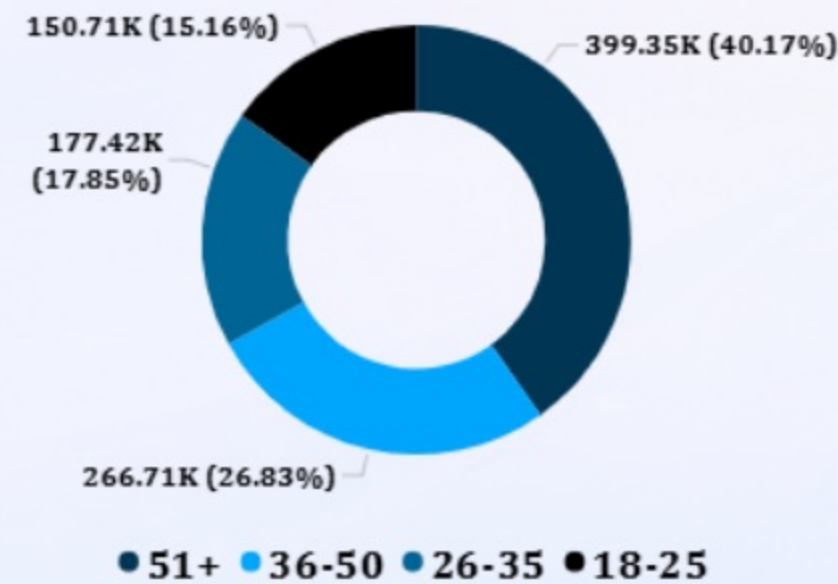
Revenue by Seasons



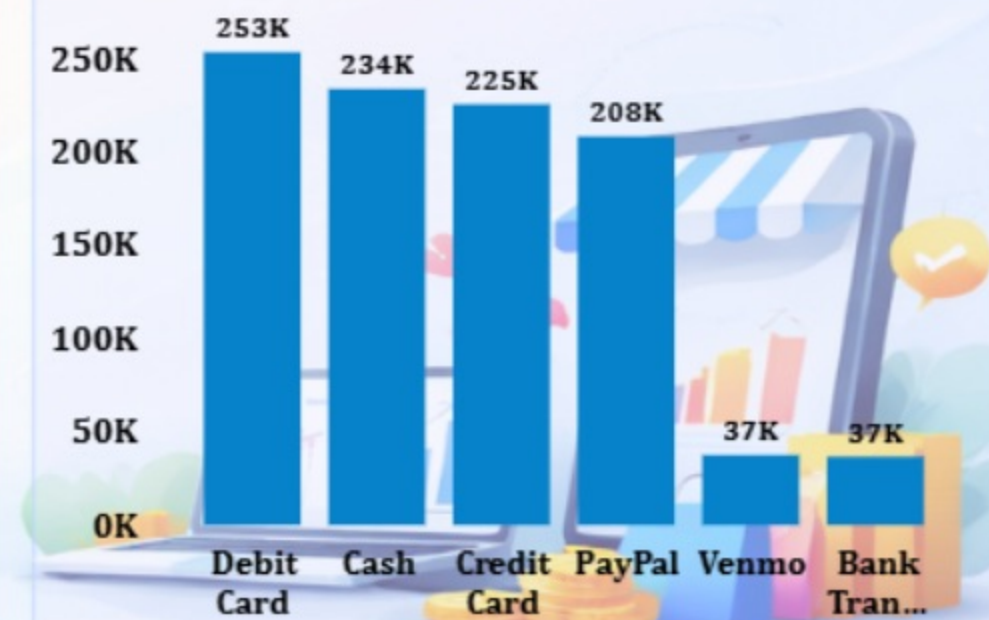
Average Revenue by Location



Revenue by Age Distribution



Revenue by payment_method



CUSTOMER BEHAVIOUR ANALYSIS

Category
All

Gender

Female

Male

Other

Discount Applied

No

Yes

Subscription Status

No

Yes

RESET

08

5K

Total_customer



30

Unique Items



4

Average Review



994.18K

Total Spend

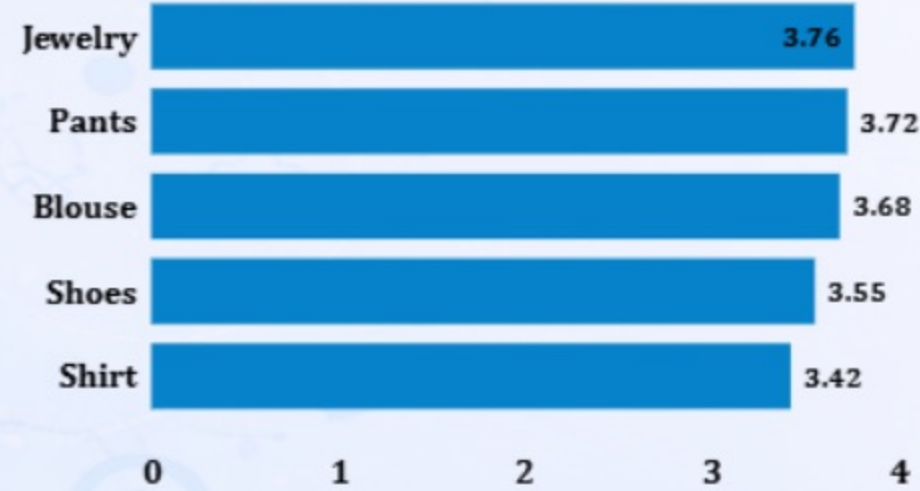


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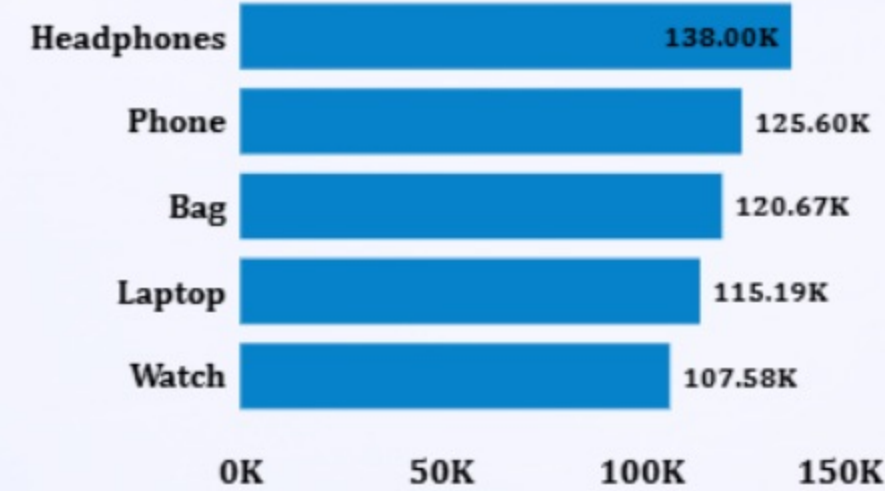
Average Spend



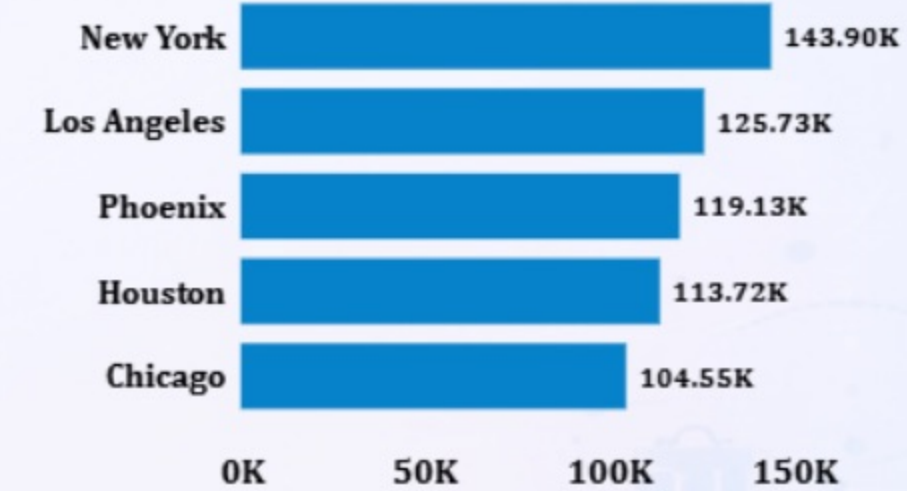
Top 5 Items by Revenue



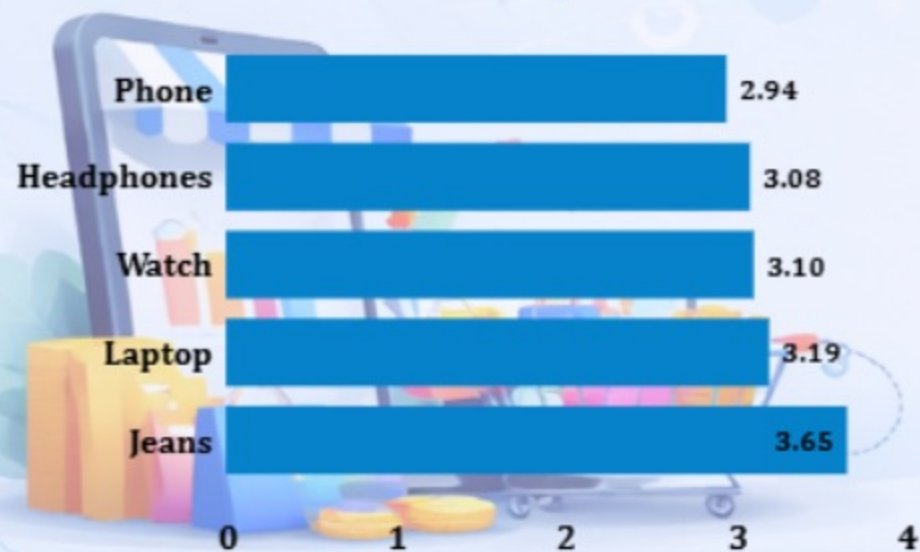
Top 5 Items by Revenue



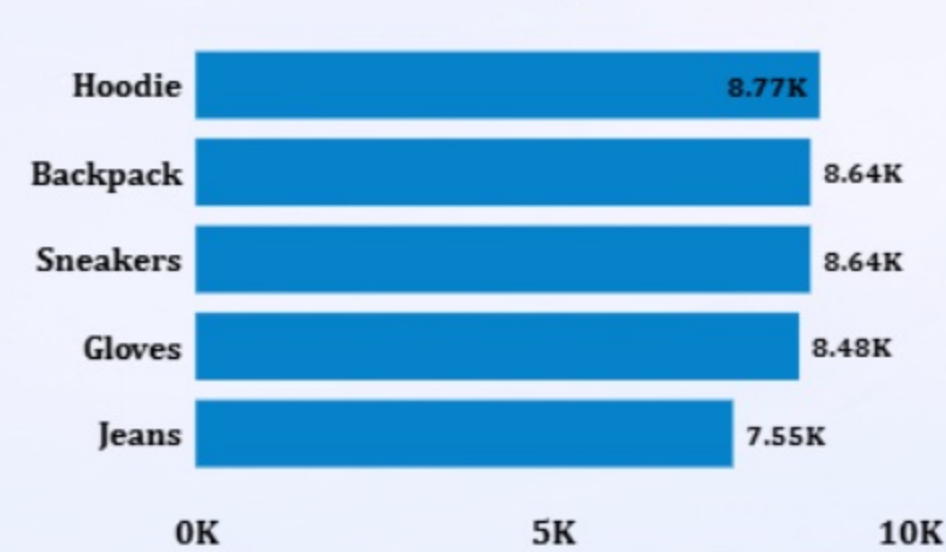
Top 5 Items by Revenue



Bottom 5 Items by Revenue



Bottom 5 Items by Revenue



Bottom 5 Items by Revenue

